LT 7501 – Computational Thinking & Human-Computer Interaction

HCI Heuristic Evaluation

Link: https://studio.code.org/

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I chose Hour of Code: Dance Party (AI Edition) for this heuristic evaluation. It is a browser-based, block-based coding activity in which learners use drag-and-drop programming to choreograph animated dancers, respond to music, and generate backgrounds using AI by selecting emojis. I worked through all 12 puzzles plus the free-build section, paying close attention to how each design decision in the platform supported or hindered learning, especially for upper elementary and middle school students encountering coding for the first time.

Assessment of All 10 Usability Heuristics

For each heuristic, I asked: *Does Dance Party (AI Edition) meet this heuristic for upper-elementary/middle-school learners? If not, how could it be improved?*

#	Heuristic	Criteria Met?	Evidence from Dance Party (AI Edition)	Improvement (if needed)
H1	Visibility of System Status	Yes	A numbered progress bar is always visible at the top. A green checkmark indicates completed puzzles. Dancers respond immediately when "Run" is clicked, so students always know whether their code is working and where they are in the activity.	
H2	Match Between System and the Real World	Yes	Block names use everyday language: "dancer does move," "when song reaches chorus," and "set background effect." Students immediately understand the dance.	
H3	User Control and Freedom	Yes	Every action can be undone. Blocks can be deleted or swapped freely. "Reset" is always visible. Students can revisit any earlier puzzle via the progress bar. No time limits and no penalties for wrong answers.	
H4	Consistency and Standards	Partial	Block color-coding and naming conventions are consistent across all puzzles.	Gradually reduce instructional detail across puzzles rather

			However, the instruction panel's level of detail drops significantly in later puzzles (Puzzles 9–10), shifting from specific guidance to vague prompts like “be creative” without warning. This can confuse students who relied on detailed instructions at earlier levels.	than dropping it abruptly. Add a visible "difficulty level" indicator so students know when they are entering a more open-ended section.
H5	Error Prevention	Partial	Block-based coding effectively prevents syntax errors. However, students can accidentally place blocks outside the valid workspace area (the gray zone), rendering them nonfunctional without any warning. There is also no autosave reminder or warning if a student tries to navigate away from an unsaved free-build project.	When blocks are dragged outside the valid workspace area, add a visual highlight or warning. When students navigate away from free-build mode, add an autosave indicator and a "you have unsaved changes" prompt.
H6	Recognition Rather Than Recall	Partial	During the guided puzzles, available blocks are clearly shown and limited to what is needed. However, in the free-build section, the full toolbox expands to show many more blocks at once, with no search bar and no "recently used" section. Students must recall which category contains which block.	Add a keyword search bar to the toolbox. In free-build mode, add a "Recently Used" section pinned to the top of the toolbox so students can quickly re-access blocks they already learned during the guided puzzles.
H7	Flexibility and Efficiency of Use	Partial	The activity offers guided and free-build modes that support different skill levels. However, there are no shortcuts, and there is no way to save custom configurations.	Allow students to save their most-used block combinations as “favorites” set to improve efficiency in free-build mode.
H8	Aesthetic and Minimalist Design	No	The landing screen is visually overwhelming: animated dancers, flashing banners, multiple call-to-action elements, artist partnerships, and social sharing buttons all compete for attention before any coding begins. Students	Simplify the entry screen to display only: a one-sentence description, a single animation preview, and a prominent "Start Coding" button.

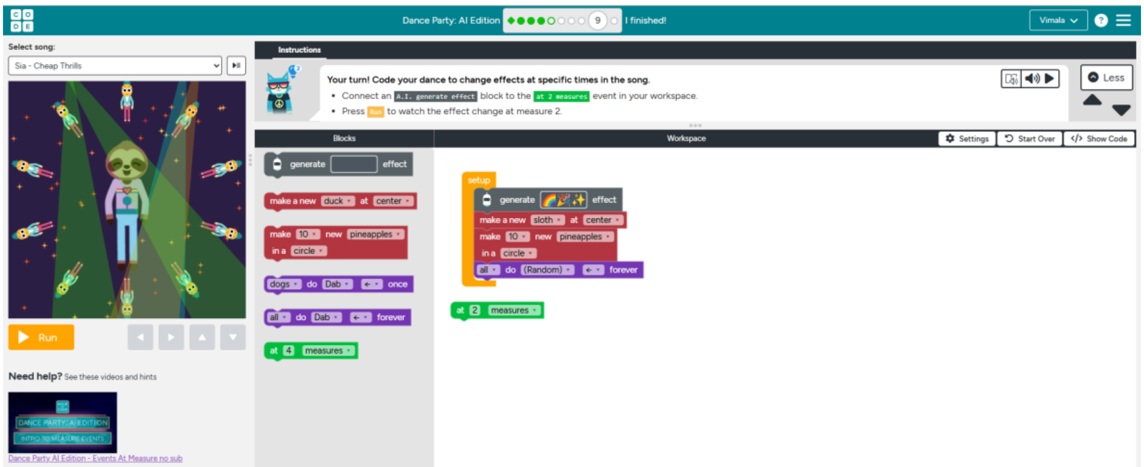
			must search for the “Start Coding” button amid the visual noise.	
H9	Help Users Recognize, Diagnose, and Recover from Errors	Yes	A lightbulb hint button is available on every guided puzzle. This is excellent. Feedback messages appear under the instructions when the code has a bug.	
H10	Help and Documentation	No	There is no in-platform reference guide, no tooltips for blocks, and no documentation explaining how the AI feature works. When the AI generates a background from emoji inputs, no explanation is provided for why that particular output was chosen.	Add hover tooltips on every block with a plain-language description. Add a collapsible "About the AI" panel explaining in simple terms how emoji embeddings work.

Heuristic Evaluation Results

3 Areas that Meet Heuristics

1. **H1 Visibility of System Status**

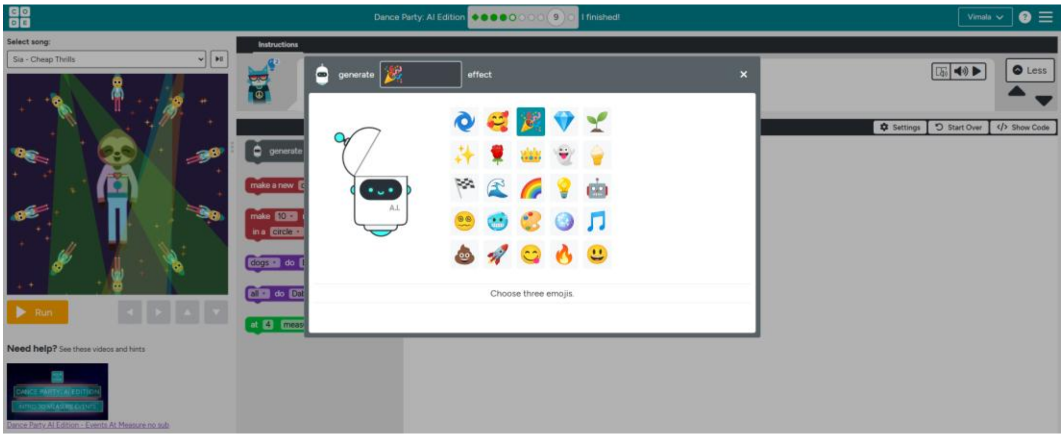
Dance Party (AI Edition) consistently keeps learners informed about what is happening at every moment. A clearly numbered progress bar at the top of the screen shows exactly which puzzle you are on (e.g., Puzzle 5 of 9), and completed puzzles are marked with a green checkmark. When you click "Run," dancers respond immediately in the play space on the right, moving, changing color, or reacting to the music in real time. Students never have to guess whether their code is working because the result is visually and audibly confirmed within a fraction of a second.



This real-time feedback loop is one of the platform's most powerful learning features. As I worked through the puzzles, I noticed that every time I clicked "Run," I instinctively looked at the play space to see whether the dancers responded as expected. That natural behavior, write code → observe result → adjust, is exactly the iterative improvement cycle. When a student places a "when song reaches chorus, dancer does move" block and immediately hears the music and sees the dancer respond, they understand the cause-and-effect relationship in a way no written definition could convey.

2. H2 Match Between System and the Real World

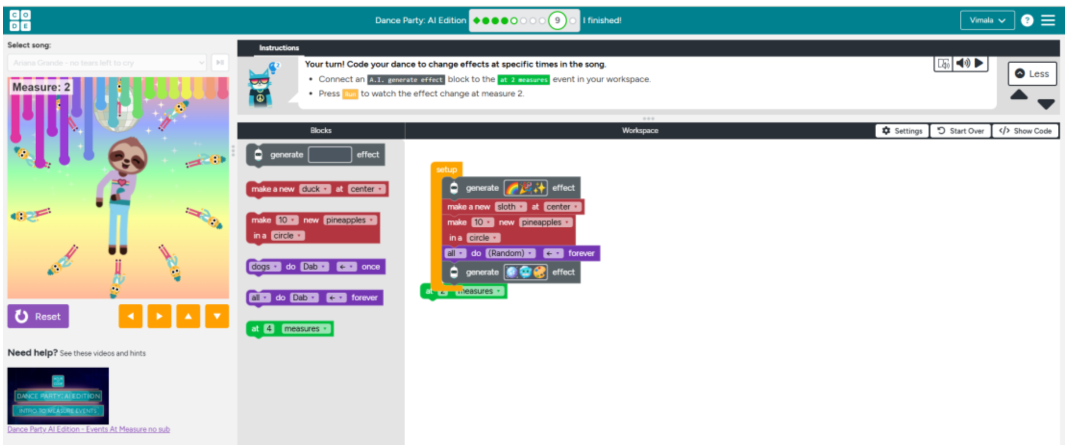
Dance Party (AI Edition) is exceptional at using language and context that students already understand. Block names are written in plain, action-based language: “dancer does move," “when song reaches chorus,” “set background effect to.” The activity is built around contemporary music from artists.



The system speaks the user’s language. When coding a dance party, students intuitively understand the goal, create something that looks and sounds great, and evaluate their own work against that standard without needing rubric. The concept of “objects-to-think-with” holds that abstract concepts become accessible when anchored in something concrete that students already care about. Music, dancing, and cultural relevance are exactly that kind of anchor. The result is that students aren’t learning about programming; they are learning programming through something that already matters to them.

3. H3 User Control and Freedom

Throughout all puzzles, Dance Party makes it completely safe to experiment. Every block placement can be undone, every dancer can be removed, and the “Reset” button always restores the workspace to its starting state without losing overall progress. Students can revisit any previously completed puzzle by clicking its dot in the progress bar. There are no time limits, no points deducted for wrong answers, and no warnings that make students feel like they are failing.

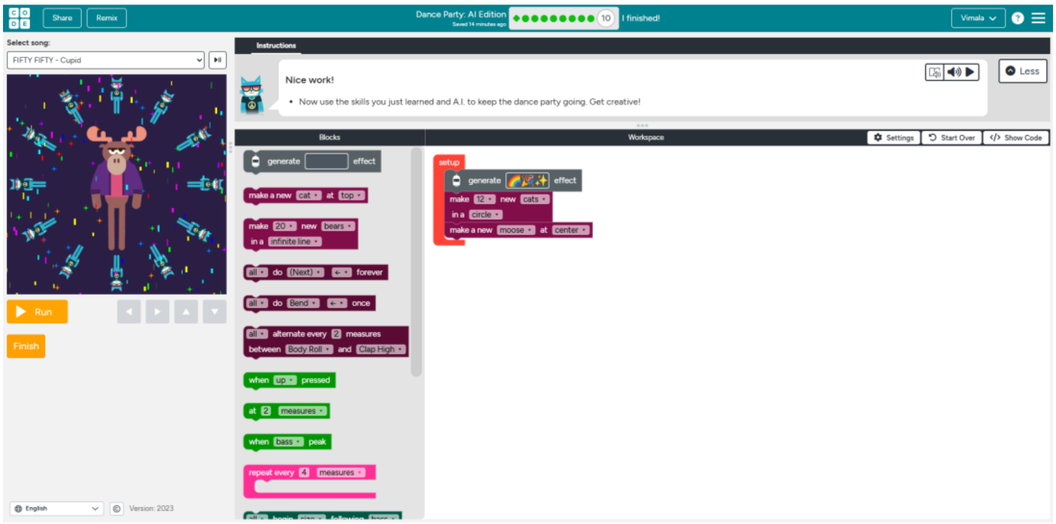


This matters deeply for the age group this activity targets. In design learning, when students develop a fear of failure, they become reluctant to try anything they are not already certain will work because mistakes feel consequential. I found myself trying unusual block combinations during free build specifically because I knew the "Reset" button would bring everything back if it didn't work. That willingness to experiment is cultivated in young learners so genuine design thinking can develop.

3 Areas for Improvement

1. **H6 Recognition Rather than Recall**

During the 10 guided puzzles, the block toolbox is beautifully managed. Only the blocks needed for each puzzle are shown, perfectly reducing cognitive load. But when students enter the free-build section (Puzzle 10 and beyond), the full toolbox expands significantly. All block categories become available simultaneously, with no search bar and no "recently used" section. This shifts the cognitive demand from creative design to menu navigation. It is always easier for users to recognize something they have seen than to recall it from memory. If students must simultaneously manage the memory burden of finding blocks, they have less mental capacity left for the actual creative and computational work. This is especially problematic for the age group targeted here. A genuine learner agency, the ability to be in control of one's own creative and intellectual process, depends on having the right tools accessible when needed. Searching through category menus is not designing; it's just navigation, and it should not be the limiting factor in what students can create.

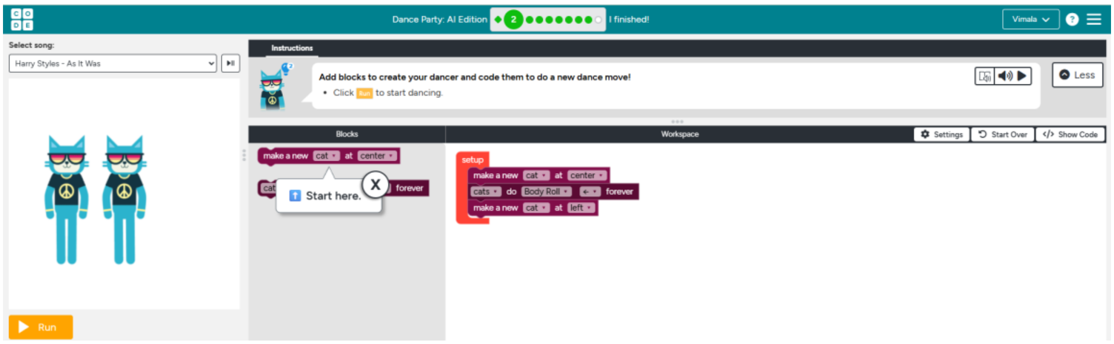


Recommendation:

Add a keyword search bar to the top of the toolbox in free-build mode so students can type "chorus" or "arrow" and immediately see all relevant blocks. Add a "Recently Used" section pinned to the top of the toolbox that shows the last 8–10 blocks placed, so students can quickly re-access tools they already learned during the guided puzzles. Together, these two changes would significantly reduce the memory burden and let students focus their energy on creating.

2. **H8 Aesthetic and Minimalist Design**

The landing page for Dance Party (AI Edition) is visually saturated. Before a student click anything, the screen displays animated dancing characters, scrolling color banners, artist partnership logos, a promotional description, social sharing buttons, a "Start" button, related activity links, and a certificate offer. While each element may have a legitimate purpose, they all compete for visual attention. On my first visit to the page, I spent time simply locating the "Start Coding" button because it was visually competing with so many other elements. For a student encountering the page for the first time, possibly already feeling nervous about coding, this kind of visual noise can feel intimidating rather than welcoming.

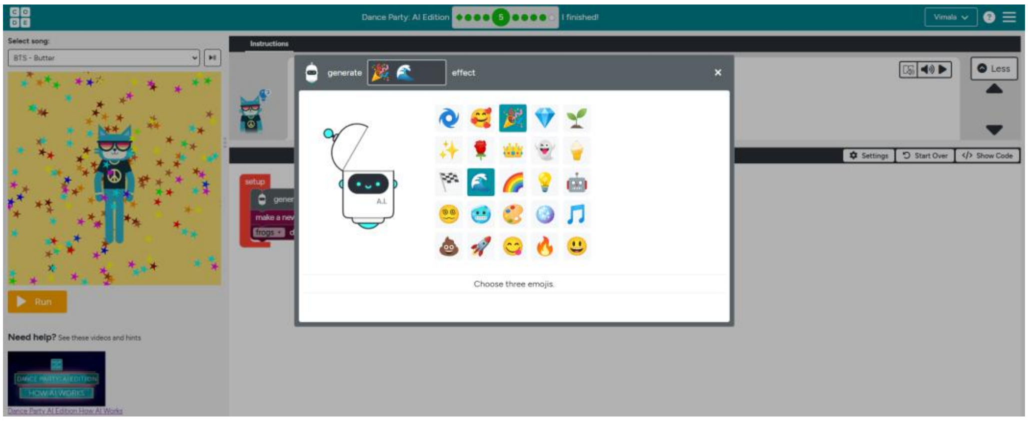


Recommendation:

Redesign the entry screen to display only three elements: (1) a single animated preview of the finished dance party, (2) a one-sentence description of what students will create ("Use code and AI to choreograph your own dance party!"), and (3) a single prominent "Start Coding" button. Move the artist branding, social sharing, and related activity links to the completion screen, where they make contextual sense as a reward after finishing — not as competition for attention at the start.

3. H10 Help and Documentation

Despite the AI feature being one of the most distinctive and educational elements of this activity, Dance Party provides no in-platform explanation of how it works. When a student selects three emojis and the AI generates a background, there is no information about what the AI did, why it chose that particular visual output, or what the word "embedding" means in this context. The AI is presented as a black box: you put emojis in, something comes out. For an activity specifically designed to introduce AI concepts to young learners, this is a significantly missed opportunity. Similarly, hovering over individual blocks in the toolbox produces no tooltips. Students who want to know what "set foreground effect" does must guess or experiment without guidance. This is a clear H10 violation. But it also represents a deeper missed learning opportunity.



Recommendation: Add hover tooltips to every block with a plain-language description and one example sentence. After the AI generates a background, show a small "How did the AI do that?" pop-up that explains in two sentences: "The AI compared the meaning of your emojis to descriptions of visual effects. It picked the one that felt most similar." Add a collapsible "About the AI" sidebar in the free-build section with a simple, age-appropriate explanation of embeddings and similarity matching.

Summary:

Dance Party (AI Edition) is among the most accessible and engaging introductory coding activities for upper elementary and middle school students. Its greatest strengths are rooted in what makes learning stick: immediate, visible feedback that shows students the consequences of every block they place; a risk-free environment that dismantles fear of failure and encourages genuine experimentation; and a real-world cultural context — contemporary music, dancing, and AI — that makes programming feel relevant and exciting rather than abstract and intimidating. The AI background feature, in particular, is a genuinely innovative design choice that introduces real machine learning concepts through a creative, emotionally resonant activity.

However, the platform has meaningful gaps that prevent it from reaching its full potential as a learning tool. The absence of any in-platform explanation of how the AI works (H10) turns what could be the most educational feature into a black box. Students see AI in action but leave without understanding what they witnessed. The visually overwhelming entry screen (H8) creates an unnecessary barrier before learning even begins. And the lack of search or recognition support in free-build mode (H6) places memory demands on students at exactly the moment their creative energy should be directed toward designing.

With targeted improvements to documentation, entry-screen simplicity, and block discoverability, it could also become excellent at supporting program-space design, giving students the tools and understanding to build confidently and independently, with a full understanding of the AI they are using. That is when an engaging activity becomes a genuinely transformative learning experience.